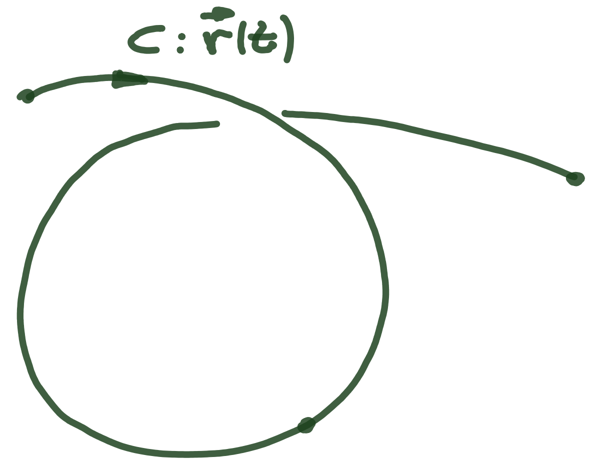
Tangent Lines and Tangent Planes



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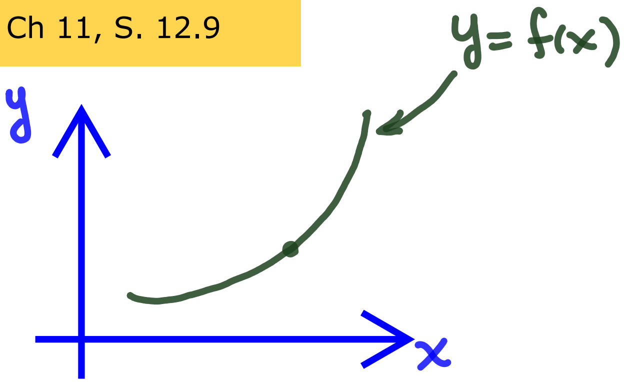
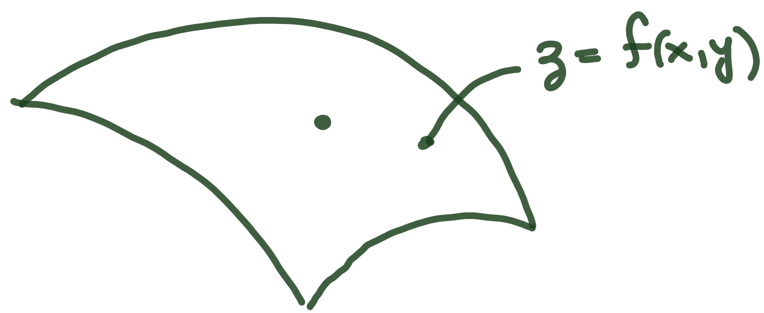
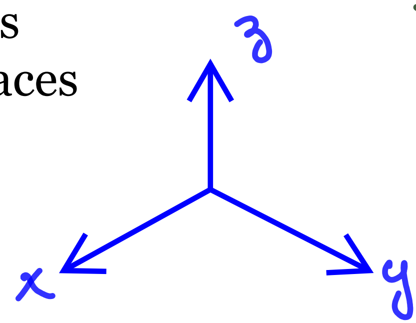


Outline:

1. Tangent Lines to 3D Curves
2. Tangent Planes to 3D Surfaces





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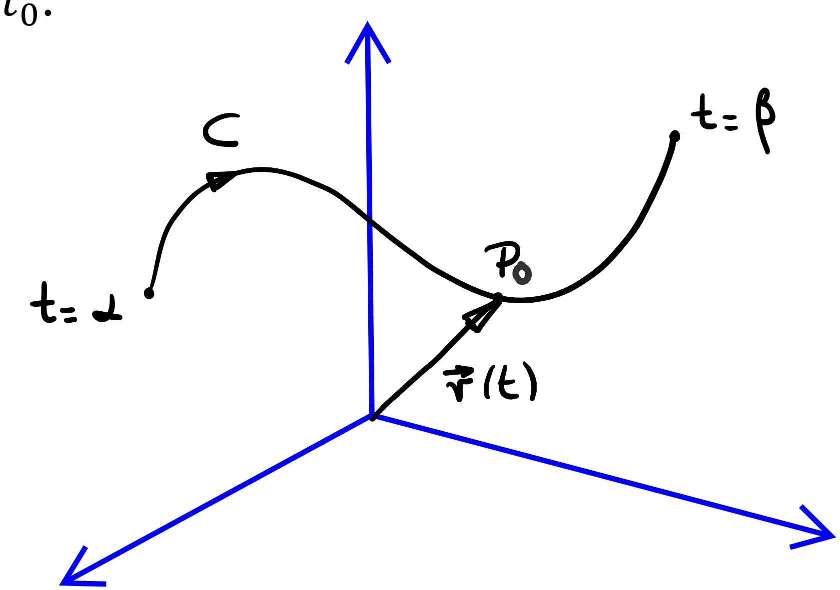
Tangent Lines to Curves

Let us assume that we have the following smooth curve C parametrized by the vector function $\vec{r}(t) = g(t)\hat{\imath} + h(t)\hat{\jmath} + k(t)\hat{k}$, $\alpha \leq t \leq \beta$, and a point $P_0(x_0, y_0, z_0)$ in that curve defined at $t = t_0$.



The equation of the tangent line to the curve C: $\vec{r}(t)$ at the point $P_0(x_0, y_0, z_0)$ can be written as,

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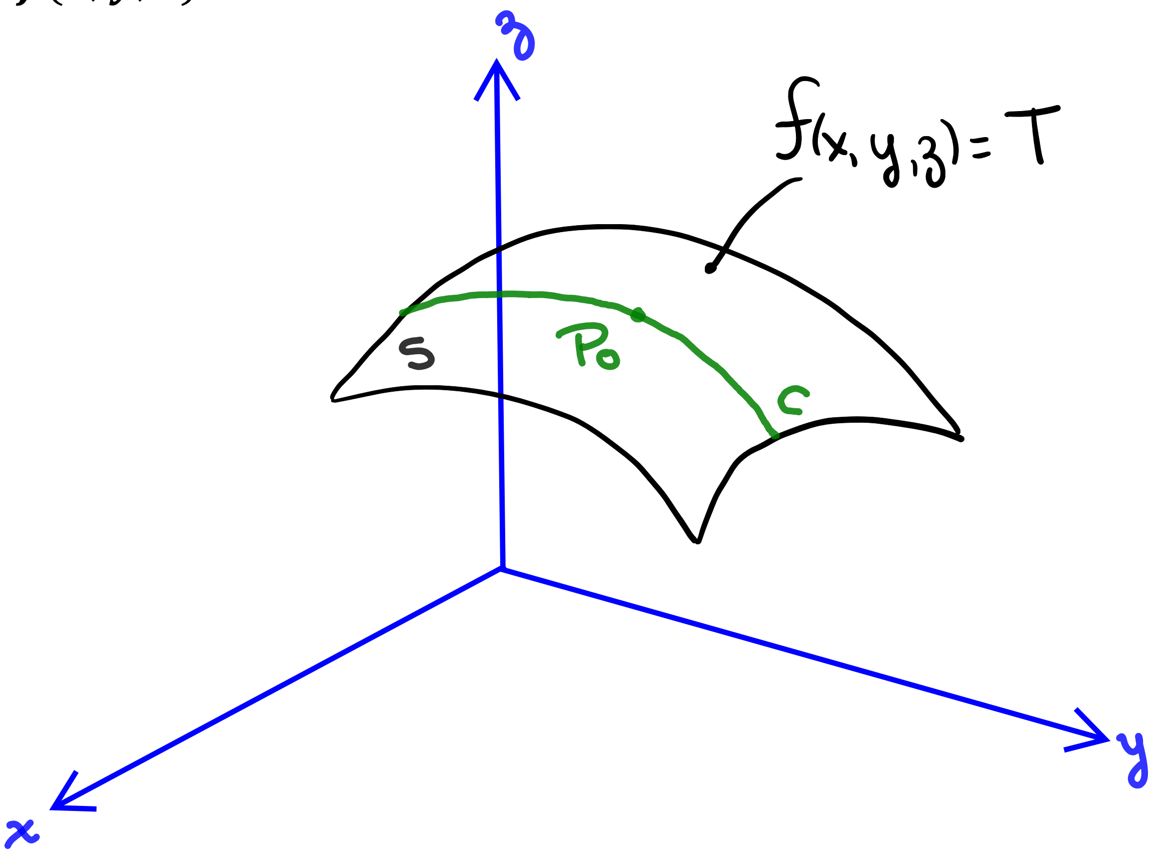
Tangent Planes to Surfaces

Using a similar approach, we can find the tangent line to a smooth curve C, which is parametrized by the vector function $\vec{r}(t) = g(t)\hat{\imath} + h(t)\hat{\jmath} + k(t)\hat{k}$, $\alpha \leq t \leq \beta$ at the point $P_0(x_0, y_0, z_0)$, where **C lies within the level surface defined by $f(x, y, z) = T$.**



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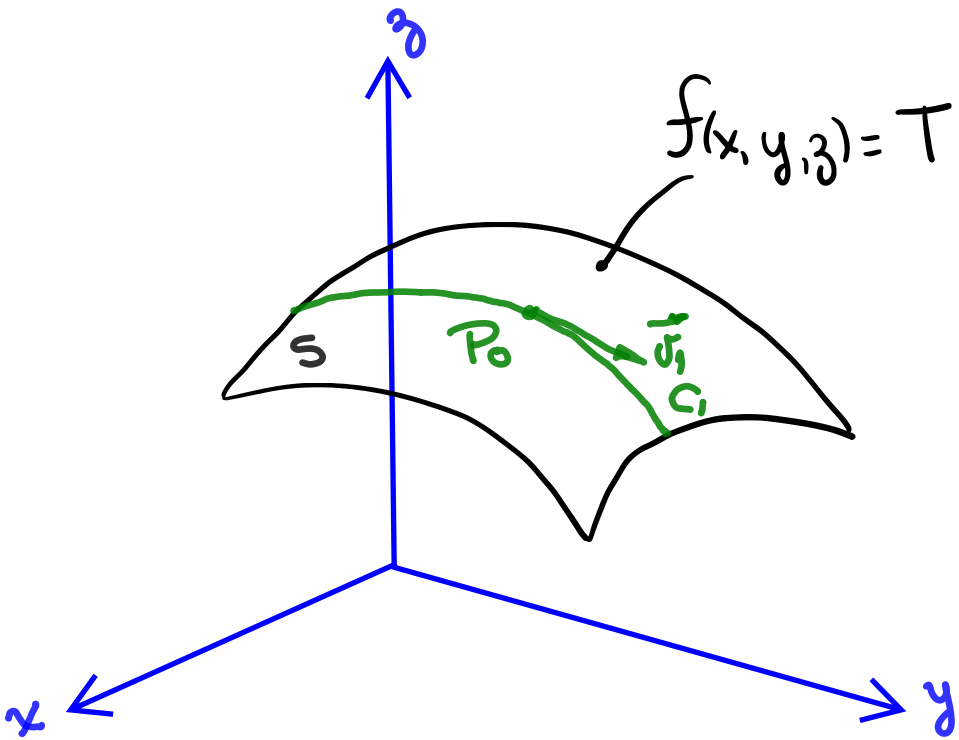
Tangent Planes to Surfaces

At every point along the curve C, $\vec{\nabla} f$ is orthogonal to $\frac{d\vec{r}}{dt}$!

Tangent Planes to Surfaces

Now, let us look at the other curves in $f(x, y, z) = T$ that pass through $P_0(x_0, y_0, z_0)$.

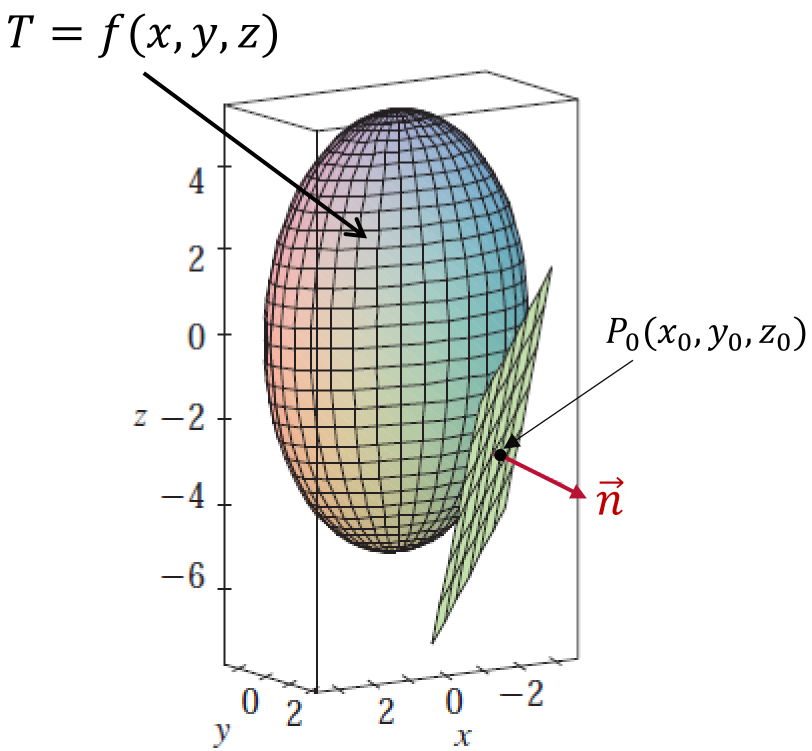


Tangent Planes to Surfaces

The normal to the tangent plane to $f(x, y, z) = T$ at P_0 is then given by,

And the tangent plane to $f(x, y, z) = T$ at P_0 is,

Normal Line to Surfaces



How would we write the equation of the normal line to the surface level $T = f(x, y, z)$ at P_0, in parametric form?

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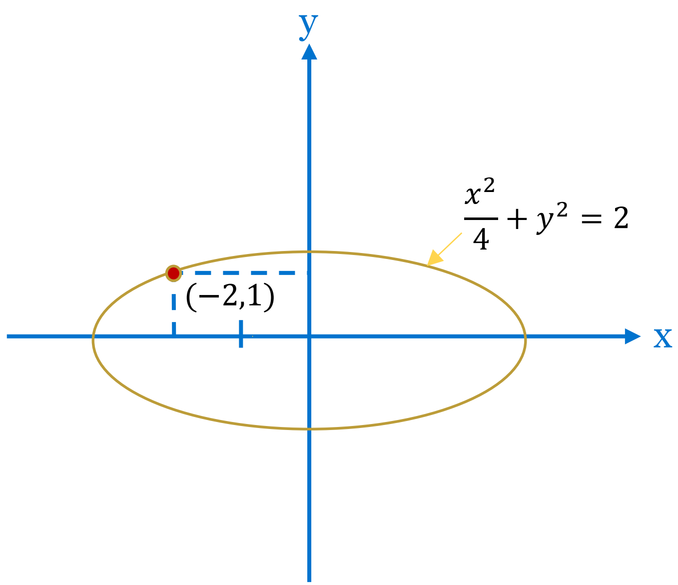
Example

Find the equation of the tangent line to the ellipse $\frac{x^2}{4} + y^2 = 2$ at the point $(-2,1)$.



Example

Summary

- The tangent line to Po in a curve defined by $\vec{r}(t)$ is nothing else than the equation of a line passing through P_0 in the direction of the velocity vector $\vec{v}(t)$

- The tangent plane to the level surface S: $f(x, y, z) = c$ at P_0 is the plane that passes through P_0 and has a normal vector equal to the gradient vector at P_0, $\left(\vec{\nabla} f\right)_{P_0}$

- The normal line to the level surface S: $f(x, y, z) = c$ at P_0 is the line passing through P_0, perpendicular to the tangent plane, and has a direction equal to the gradient vector at P_0 , $\left(\vec{\nabla} f\right)_{P_0}$

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